

Decentralized Health Management System (DHMS)

1. Title and Brief Summary of the Initiative/Project:

The Decentralized Health Management System (DHMS) is a blockchain-based solution developed to securely manage and share patient health records among healthcare providers. The system ensures data privacy, transparency, and control through a decentralized architecture that removes the need for a central authority.

Our project was presented at the Blockcathon 2025, held at Sri Eshwar College of Engineering, Coimbatore, where it received 3rd Prize among several innovative blockchain-based projects.

2. Goal and Problem Addressed:

- The goal of this project was to **revolutionize healthcare data management** by eliminating data silos and central dependencies through blockchain technology. Existing systems often suffer from **security breaches, lack of interoperability, and limited patient control**. DHMS addresses these problems by allowing patients to **own and control access** to their medical data using **smart contracts and cryptographic verification**.
- By leveraging **Quorum blockchain**, the project ensures **permissioned access, encrypted record storage, and tamper-proof medical data management**, allowing patients to control how and when their data is shared.

3. Number of Students Involved and Their Roles:

Total Students Involved: 4

Name	Role	Contributions
Nihil	Project Lead & Blockchain Developer	Designed and deployed smart contracts on the Quorum blockchain; managed decentralized architecture.
Rahith	Backend Developer	Built APIs and database integrations to connect blockchain with user interfaces.
Monish	Frontend Developer	Developed the web interface for both patients and healthcare providers.
Syed	Data & Security Analyst	Ensured data privacy through encryption techniques and optimized data storage.

4. Key Activities and Outcomes Achieved:

- Participated in **Blockcathon 2025** at **Sri Eshwar College, Coimbatore**.
- Designed and implemented **smart contracts** for secure data transactions.
- Integrated **IPFS (InterPlanetary File System)** for decentralized storage of patient records.
- Conducted **live demos** showcasing record creation, verification, and access permissions.

Outcomes:

- ✔ Successfully demonstrated a working prototype.
- ✔ Achieved **secure and transparent health record sharing**.
- ✔ Received **3rd Prize** for innovation and technical execution.

5. Is this Initiative a One-Time Event or Part of a Longer Program?

The participation at Blockcathon was a **single event**, but the project itself is part of a **long-term research initiative** focusing on **blockchain applications in healthcare**. Future goals include integrating **AI for predictive health analytics** and **collaborating with healthcare institutions** for real-world pilot testing.

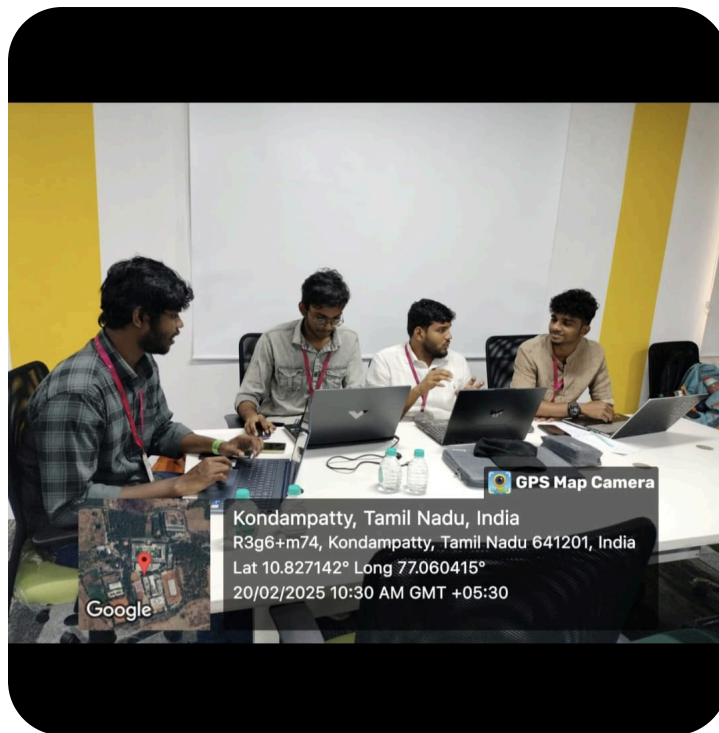
6. New or Creative Approaches Used in the Initiative:

- Developed a **patient-controlled data access system** using **Ethereum smart contracts**.
- Used **IPFS** for off-chain storage of large health files while maintaining blockchain verification.
- Implemented **Role-Based Access Control (RBAC)** for secure data permission handling.
- Combined **decentralized data management** with a **user-friendly web interface** for accessibility.

7. Challenges Faced and How They Were Overcome:

Challenge	Solution Implemented
Integrating blockchain with conventional data systems	Created middleware APIs for seamless data flow
Protecting sensitive health data	Used encryption and stored only metadata on-chain
Managing real-time data verification during demo	Simulated test environments using Ganache and MetaMask
Time constraint during Blockcathon	Optimized development workflow and prioritized key functionalities

8. Supporting Materials



Testimonials:

“An innovative application of permissioned blockchain technology in healthcare. Demonstrated strong technical depth and real-world relevance.” — *Blockcathon Judge Panel, Sri Eshwar College*

